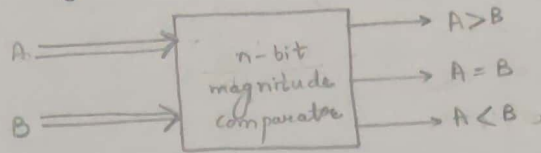


Magnitude Comparator

A Magnitude Comparator is a combinational circuit that compares the magnitude of two numbers (A and B) and generates one of the following outputs :- $A=B$, $A<B$ & $A>B$

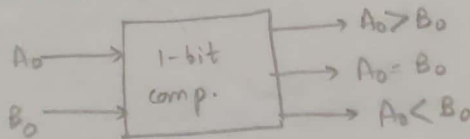


Block diagram of n-bit comparator

The X-NOR gate (coincidence gate) is a basic comparator, because its o/p is a 1 only if its two i/p bits are equal, i.e. the o/p is a 1 if and only if the i/p bits coincide.

⇒ To implement the magnitude comparator, the X-NOR & AND gates are used.

1-bit magnitude comparator -



A_0	B_0	$A_0 > B_0$	$A_0 = B_0$	$A_0 < B_0$
0	0	0	1	0
0	1	0	0	1
1	0	1	0	0
1	1	0	1	0

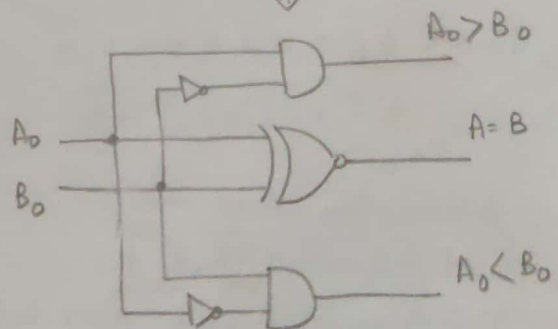
⇒

$$A_0 > B_0 = A_0 \bar{B}_0$$

$$A_0 = B_0 = A_0 \odot B_0$$

$$A_0 < B_0 = \bar{A}_0 \cdot B_0$$

⇓



Logic diagram

2-bit Magnitude Comparator

Inputs				Outputs		
A ₁	A ₀	B ₁	B ₀	A > B	A = B	A < B
0	0	0	0	0	1	0
0	0	0	1	0	0	1
0	0	1	0	0	0	1
0	0	1	1	0	0	1
0	1	0	0	1	0	0
0	1	0	1	0	1	0
0	1	1	0	0	0	1
0	1	1	1	0	1	0
1	0	0	0	1	0	0
1	0	0	1	1	0	0
1	0	1	0	0	0	1
1	0	1	1	0	0	1
1	1	0	0	1	0	0
1	1	0	1	1	0	0
1	1	1	0	1	0	0
1	1	1	1	0	1	0

A ₁ A ₀ \ B ₁ B ₀	00	01	11	10
00	0	1	3	2
01	4	5	7	6
11	12	13	15	14
10	8	9	11	10

$$\begin{aligned}
 A > B &= A_1 \bar{B}_1 + A_0 \bar{B}_1 \bar{B}_0 + A_1 A_0 \bar{B}_0 \\
 &= A_1 \bar{B}_1 + (\bar{B}_1 + A_1) A_0 \bar{B}_0
 \end{aligned}$$

∴ conditions for $A > B$ —

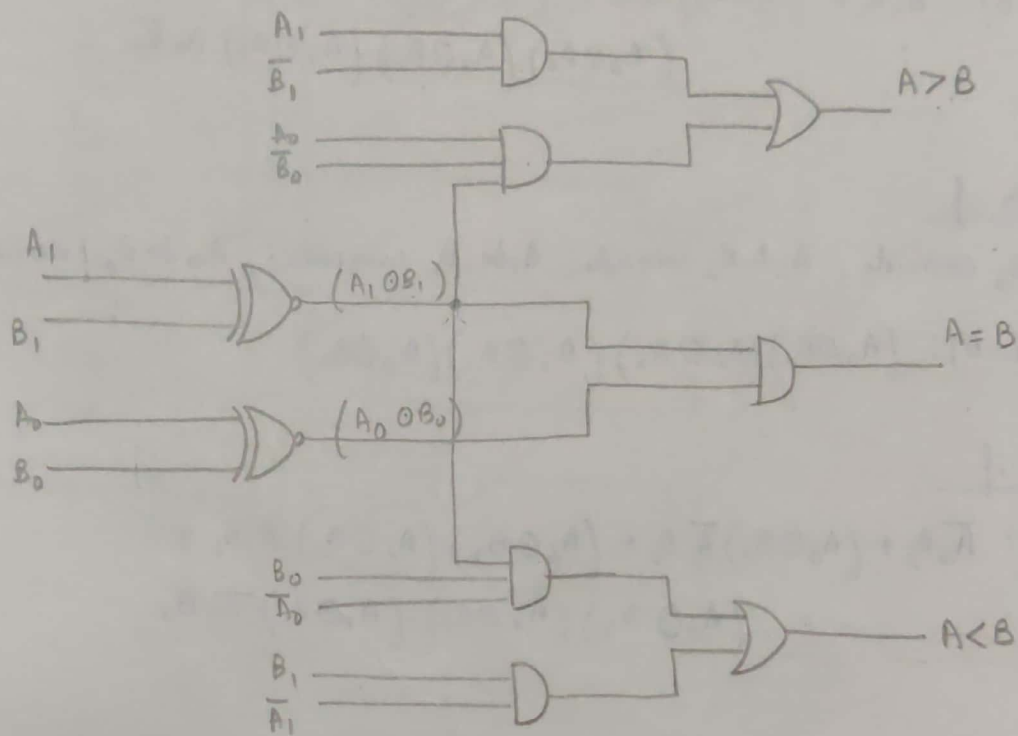
$$\left. \begin{aligned}
 &\text{i) } A_1 = 1 \text{ \& } B_1 = 0 \Rightarrow \boxed{A_1 \bar{B}_1} \\
 &\text{ii) } A_1 \text{ \& } B_1 \text{ coincide and } A_0 = 1 \text{ \& } B_0 = 0 \Rightarrow (A_1 \odot B_1) A_0 \bar{B}_0
 \end{aligned} \right\} \Rightarrow A_1 \bar{B}_1 + (A_1 \odot B_1) A_0 \bar{B}_0$$

conditions for $A = B$ -

- i) A_1 & B_1 coincide & A_0 & B_0 coincide
 $(A_1 \odot B_1)(A_0 \odot B_0)$

conditions for $A < B$ -

- i) $A_1 = 0$ & $B_1 = 1 \Rightarrow \bar{A}_1 B_1$
 ii) A_1 & B_1 coincide & $A_0 = 0$ & $B_0 = 1 \Rightarrow (A_1 \odot B_1) \bar{A}_0 B_0$
 $(A_1 \odot B_1) \bar{A}_0 B_0$



Logic diagram of 2-bit Magnitude Comparator

4-bit Magnitude Comparator -

Let two 4-bit numbers be $A = A_3A_2A_1A_0$ & $B = B_3B_2B_1B_0$.

$A > B$ if -

i) $A_3 = 1$ & $B_3 = 0 \Rightarrow A_3 \bar{B}_3$

ii) A_3 & B_3 coincide and $A_2 = 1$ & $B_2 = 0 \Rightarrow (A_3 \odot B_3) A_2 \bar{B}_2$
coincide

iii) A_3 & B_3 coincide, A_2 & B_2 coincide and $A_1 = 1$ & $B_1 = 0$
 $\Rightarrow (A_3 \odot B_3) (A_2 \odot B_2) A_1 \bar{B}_1$

iv) A_3 & B_3 coincide, A_2 & B_2 coincide, A_1 & B_1 coincide, $A_0 = 1$ & $B_0 = 0$.
 $\Rightarrow (A_3 \odot B_3) (A_2 \odot B_2) (A_1 \odot B_1) A_0 \bar{B}_0$

$\therefore (A > B): A_3 \bar{B}_3 + (A_3 \odot B_3) A_2 \bar{B}_2 + (A_3 \odot B_3) (A_2 \odot B_2) A_1 \bar{B}_1 +$
 $(A_3 \odot B_3) (A_2 \odot B_2) (A_1 \odot B_1) A_0 \bar{B}_0$

$A = B$ if

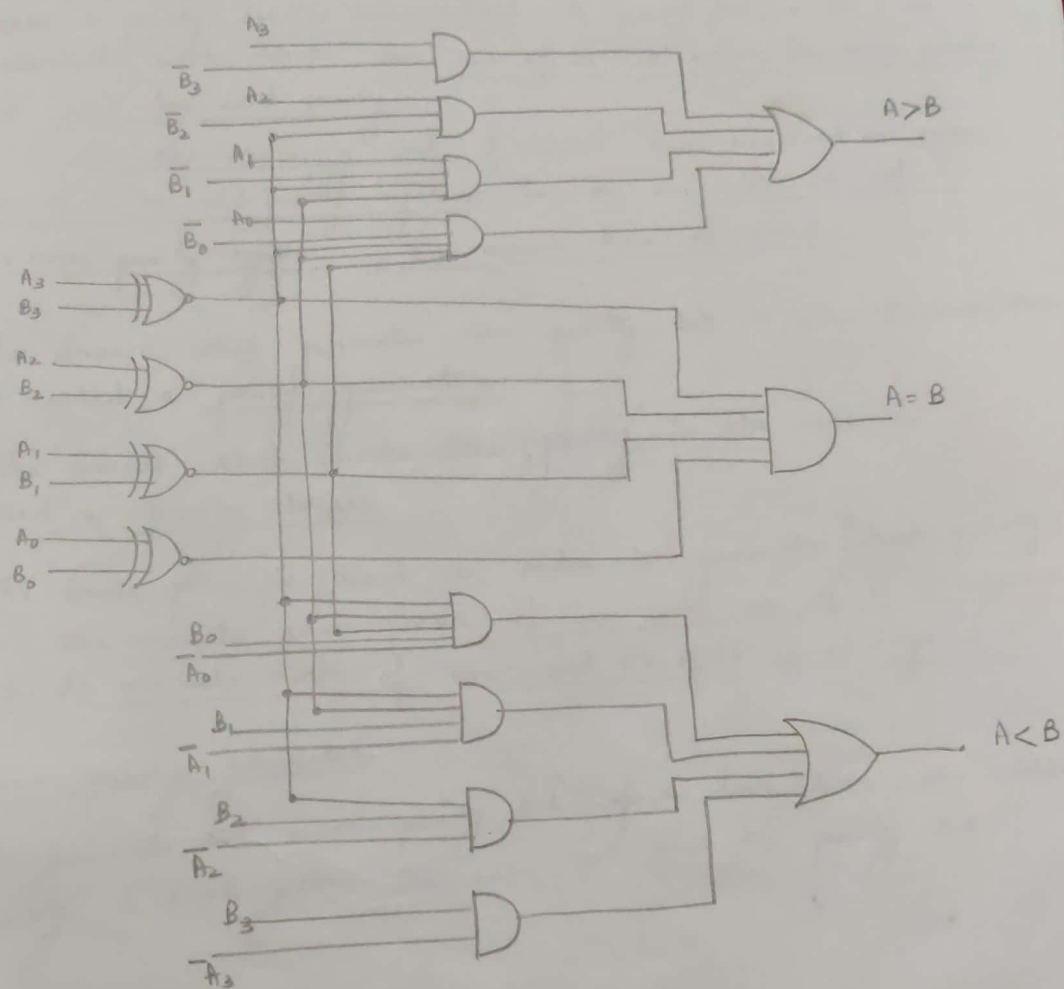
A_3 & B_3 coincide, A_2 & B_2 coincide, A_1 & B_1 coincide, A_0 & B_0 coincide

$(A = B): (A_3 \odot B_3) (A_2 \odot B_2) (A_1 \odot B_1) (A_0 \odot B_0)$

$A < B$ if

$(A < B): \bar{A}_3 B_3 + (A_3 \odot B_3) \bar{A}_2 B_2 + (A_3 \odot B_3) (A_2 \odot B_2) \bar{A}_1 B_1 +$
 $(A_3 \odot B_3) (A_2 \odot B_2) (A_1 \odot B_1) \bar{A}_0 B_0$

The logic diagram of a comparator is shown next:-



Logic diagram of a 4-bit Magnitude Comparator